

Abstract

Automated fault-injection + 0-100 Resilience Index $RI=(D \times 0.4)+(Rec \times 0.3)+(S \times 0.3)$ + rule-based diagnosis + PyQt6 GUI vs HIL \$50K. Tested on STM32F4/nRF52840/RISC-V with 27 faults, achieving Grade B (72 to 78) after fixes.

1. Introduction

Recall problem \$500M, HIL cost, no metric, Renode has no structured testing layer (README.md:32).

2. Related Work

QEMU custom vs Vector vs Renode - Table README.md:745.

3. Fault Taxonomy

27 types 6cats SF-01..GF-02, Renode hooks sysbus WriteDoubleWord, verified renode/platforms/cpus/stm32f4.repl:31 can1, i2c1, gpio.

4. Architecture

Fig 01-ARCHITECTURE.md:7 GUI(QMainWindow 1400x900)->Bridge(QProcess :1234)->Renode->ELF; QThreadPool 4, adr/002 hybrid pythonnet fallback.

5. Resilience Index

Formula 08-RESILIENCE_INDEX.md:3, weights adr/003 40/30/30, grades A90 B70 C50 D30.

6. Diagnosis

Rules 09-DIAGNOSIS: SF-03 median filter, TF-01 watchdog 50ms, threshold 50Hz, ISO26262-6.

7. Implementation

PyQt6 6.6, FastAPI, Typer, 8 core modules src/core/*, 27 fault catalog, renode 1.16.1, Docker.

8. Evaluation

3 firmwares src/main.c, campaigns full_27.yaml RI 72->78 (+8.3%) after fixes, throughput 100/hr, report <2s, pytest 23 tests 67% src.

9. Future

v1.1 ESP32 ML scikit-learn plugin, v2.0 distributed cloud, v3.0 LLM fixes (CHANGELOG).

10. Conclusion

First quantitative resilience framework on Renode, open MIT, desktop-native for embedded engineers.